

ACM Multimedia 2023 Grand Challenge Report: Invisible Video Watermark

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ABSTRACT

MGTV recently organized a pioneering Invisible Video Watermark Challenge, inviting participants to create a framework capable of embedding invisible watermarks into videos and extracting them from watermarked content.

The invisible watermark serves as a discrete digital signature within the media data, imperceptible to the human eye. This technique safeguards the ownership and authenticity of multimedia content. While convolutional neural networks have demonstrated remarkable efficacy in image and video processing, the discourse on invisible watermarking remains limited. This challenge, therefore, presents an opportune moment to advance the field of invisible watermarking.

Furthermore, to support this endeavor, we curated the comprehensive MGTV_WM dataset, encompassing diverse video types. For further details, please refer to our official website (<https://challenge.ai.mgtv.com/#/track/18?locale=en>).

CCS CONCEPTS

• General and reference → General conference proceedings.

KEYWORDS

invisible watermarking, convolutional neural networks, dataset

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1 INTRODUCTION

Invisible watermarking serves as a powerful strategy for embedding concealed information within various multimedia contents, including images, audio [2], and videos [6]. The key characteristic

of invisible watermarking is its ability to embed the watermark in a way that preserves the quality and appearance of the media, while still requiring sophisticated algorithms for extraction.

The typical process of applying an invisible watermark involves two fundamental steps [5]: embedding and extraction. Embedding refers to the initial stage of inserting the invisible watermark into the media, ensuring it remains imperceptible. Extraction, on the other hand, focuses on the subsequent process of retrieving the original watermark from the watermarked media.

Normally, transform domain methods have been widely adopted for embedding invisible watermarks into videos. However, with the advancements in deep neural networks [4], new approaches have emerged to address this challenge. Deep neural networks have demonstrated remarkable capabilities in handling image and video-related tasks, offering promising avenues for reliable and efficient invisible watermarking techniques. Some common approaches include:

- Transform Domain Methods: These methods involve applying mathematical transforms, such as Discrete Cosine Transform (DCT) [3] or Discrete Wavelet Transform (DWT) [8], to the video frames. The watermark is embedded by modifying the transform coefficients in a way that is imperceptible to the viewer.
- Spatial Domain Methods: These methods operate directly on the pixel values of the video frames. The watermark is embedded by slightly modifying the pixel values in specific regions or using techniques like Least Significant Bit (LSB) [7] substitution, where the least significant bits of the pixels are replaced with the watermark data.
- Frequency Domain Methods: These methods involve transforming the video frames into the frequency domain using techniques like Fast Fourier Transform (FFT) [12]. The watermark is embedded by modifying the amplitudes or phases of certain frequency components.
- Deep Learning Methods: Deep learning models can adapt watermark embedding strategies based on video content characteristics. They learn to identify regions suitable for embedding based on the content, resulting in improved robustness and imperceptibility.

Assessing the effectiveness of an invisible watermarking method typically revolves around its robustness against various attacks. Watermarking attacks aim to remove, modify, or destroy the embedded watermark in digital media. Some common types of watermarking

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attacks include removing watermarks entirely, leaving no trace of their existence, and modifying watermarks to diminish their effectiveness by altering their size or content. By evaluating a method's resistance to such attacks, its reliability and suitability for practical applications can be determined.

Furthermore, a few academic conferences have explored invisible watermarks, and there is no standard dataset for this purpose. Therefore, it would be challenging to assess new accomplishments on tasks with an invisible watermark. To address this issue, we have organized these challenges and provided standardized datasets. By doing so, we aim to expedite the development of invisible watermarking methods and facilitate future research in this area.

2 CHALLENGE DESCRIPTION

We provide the following resources to participants in order to design a robust, effective, and high-performance invisible video watermark solution:

- A high-quality dataset with a variety of categories for training and evaluating the solution;
- The baseline code with data processing and evaluation can help participants save time and concentrate on developing embedding and extraction algorithms;
- A set of metrics for measuring the robustness, effectiveness, and efficiency of a developed solution;
- A necessary requirement for running time and video quality, which can ensure the usability of the approaches.

2.1 Problem Definition

The central objective of this challenge is to implement an invisible video watermark using a binary watermarking approach. This binary watermark comprises data represented as a sequence of zeros and ones. The key challenge is to embed this binary watermark into the video in a manner that remains imperceptible to viewers' eyes while ensuring robust and reliable detection and extraction when needed.

The challenge in implementing an invisible watermark for a video lies in finding a balance between the watermark's robustness, imperceptibility, and resistance to removal or tampering. The watermark should be invisible to the viewer, ensuring that it does not degrade the visual quality or distract from the intended content of the video. At the same time, it should be difficult to remove or alter the watermark without leaving visible artifacts or compromising the video's integrity. To successfully implement an invisible watermark for a video, it is crucial to consider factors such as robustness against various attacks (e.g., compression, cropping, filtering), capacity (how much information can be embedded), and computational efficiency (to enable real-time or fast processing).

2.2 Dataset

Currently, there is no standard dataset to measure the invisible watermark algorithms. Various public datasets are used to measure the watermarking, such as COCO [10], CIFAR-10 [11], and CelebA dataset [1], which are widely used in detection and classification. Therefore, a standard dataset for watermarking is desideratum.

To support this challenge and further research, we build a dataset named MGTW_WM. It contains 1000 video clips, and each clip lasts

for about 30 seconds, and the resolution is 720x1280 or 480x848. Moreover, MGTW_WM videos are available in a variety of types, including animation, news, shows, documentaries, and movies. Extensive video types can better measure the robustness of watermarking.

Compared to existing datasets used in invisible watermarking, our dataset offers the following advantages:

- **Video-level:** Unlike the existing dataset used in research, MGTW_WM consists of videos, which could be used both at frame-level or video-level.
- **Large amount:** MGTW_WM contains 1000 videos, which is sufficient for study purpose.
- **Various types:** MGTW_WM contains various video types, which are close to the real production environment.
- **Independent copyright:** We hold independent copyright for the videos, which ensures the legality of the dataset for research and commercial use.

3 CHALLENGE EVALUATION

3.1 Data Setting

The dataset is divided into two parts: the public set and the testing set, according to a ratio of 9:1. The public set was released at the beginning of the challenge, which was used by participants to develop their methods, and the testing set was released after the end of the challenge, which was used to examine the ranking of participants.

3.2 Attacks

This competition did not use a single attack method but instead used several attack combinations. On one hand, when considering individual attacks, each attack applied to a video generates a separate attacked version. With a greater number of attack methods, there will be a corresponding increase in the number of attacked videos derived from the same original video. Consequently, extracting the watermark and evaluating the algorithm becomes a time-consuming process. On the other hand, in real-world scenarios, watermarks are commonly subjected to multiple attacks simultaneously. Hence, employing combination attacks offers a more realistic and efficient approach to assess the robustness of watermarking algorithms.

The challenge incorporated a total of 6 combined attacks, each designed to test the robustness of the video watermarking system. These combinations were as follows:

- Combination 1: Included a noise attack along with a resize attack.
- Combination 2: Consisted of a crop attack, manipulating video quality through bitrate adjustments and frame rate changes.
- Combination 3: Involved the addition of multiple visible watermarks to the video, alongside bitrate adjustments and frame rate changes to degrade video quality.
- Combination 4: Implemented a picture-in-picture attack, shrinking the attacked video and overlaying it on top of another video. Additionally, it affected video quality by adjusting the bitrates.

- Combination 5: Simultaneously adjusted video contrast, brightness, and saturation.
- Combination 6: Utilized a rotation attack and Gaussian noise.

Furthermore, all videos that have been attacked would also be cut at random. By employing these diverse combinations, the challenge aimed to simulate real-world scenarios and assess the effectiveness of the watermarking algorithm against multiple attack vectors.

3.3 Metric

When evaluating watermarking techniques, we take into consideration three key aspects: extraction accuracy, embedding speed, and the quality of the watermarked video.

Extraction accuracy refers to the ability of the algorithm to accurately and reliably extract the embedded watermark from the watermarked video. This aspect focuses on the robustness of the algorithm in various scenarios, including when the video undergoes common processing operations or when subjected to intentional attacks.

Embedding speed measures the efficiency of the algorithm in terms of how quickly it can embed the watermark into the video. A faster embedding speed is desirable as it allows for efficient processing and minimizes any potential delay.

The quality of the watermarked video is an essential consideration as it directly impacts the visual perception of the content. The watermark should be imperceptible to viewers, ensuring that it does not introduce noticeable distortions or artifacts that compromise the overall quality or viewing experience.

By evaluating these three aspects—extraction accuracy, embedding speed, and watermarked video quality—we can gain valuable insights into the performance and suitability of the watermarking technique for different applications and scenarios.

3.3.1 Extraction Accuracy. We adopt 256-bit as the watermarks for each frame, and the bit accuracy ratio (BAR) is used to measure the accuracy of extracted watermarks.

$$\text{BAR} = \frac{1}{X} \sum_{1 \leq i \leq X} D(WM_O(i), WM_E(i)) \quad (1)$$

$$D(x, y) = \begin{cases} 1 & x == y \\ 0 & x \neq y \end{cases}$$

where X represents the length of the watermark, WM_O represents the original watermark, and WM_E represents extracted watermark. The BAR of the video is the average of all frames' BAR, and the BAR of the whole dataset is the average of all videos' BAR.

3.3.2 Embedding Speed. The number of frames that can be embedded per second is used to measure the speed of watermarking, and it can be called frames per second (FPS). We will give several intervals to correspond to different scores. Detailed scores are shown in Table 1.

3.3.3 Watermarked Video Quality. The peak signal to noise ratio (PSNR) is to be considered when measuring the quality of watermarked videos. Watermarked videos should be of comparable quality to the original. The PSNR for a single frame is computed as follows:

Table 1: The score of embedding speed

embedding speed	score
> 40fps	0.01
15fps-40fps	0.0
< 15fps	The submitted method is invalid

Table 2: ACM MM 2023 IVW Challenge Final Results

Rank	Team name	BAR	Embed FPS	PSNR	Score
1	CB	0.8841	56	43.2	0.8941
2	xiaotu	0.8611	50	41.7	0.8711
3	mg12345678	0.862	21	41.3	0.862
4	bvc	0.7729	17	40.8	0.7729
5	shenme	0.7356	32	39.2	0.7356
6	xianlu	0.7285	31	39.1	0.7285
7	mangguo	0.7041	17	40.7	0.7041
8	mg731755782	0.6374	72	41.5	0.6474
9	huanmie	0.5938	69	41.2	0.6038
10	mg923174UzB	0.5885	48	41.0	0.5985
-	baseline	0.5073	18	41.5	0.5073

$$\text{PSNR} = 10 \log_{10} \frac{\text{MAX}_F^2}{\text{MSE}} \quad (2)$$

where MAX_F represents the maximum pixel value of the frame, and MSE is the mean square difference between the origin frame and the watermarked frame.

A video's PSNR score is calculated as the average of all its frames' PSNR, and the final PSNR is calculated as the average of all videos. In addition, we set a lower limit for the final PSNR. If the PSNR of a submission falls below the threshold value (39 dB), the submission is invalid.

The BAR and speed scores are added to produce the final result. Whenever two or more submissions have the same score, the submission with the highest PSNR will be ranked higher.

4 CHALLENGE RESULTS

A total of 1,804 participants registered for the challenge. This demonstrates that the invisible watermark is attractive. However, only 29 teams made at least one valid and executable submission. This indicates that the challenge is difficult. Most participants have no better methods after examining the baseline. Each team can only submit once each day for this challenge due to restricted computing resources and a long execution time. These 29 teams submitted 298 submissions in total. Table 2 summarizes the final challenge results. Team CB shows the best overall performance, followed by Team xiaotu and Team mg12345678.

5 CHALLENGE METHODS

In this section, we describe some solutions that ranked in the top 10 of the ACM MM 2023 IVW Challenge. Six of the top ten teams employed deep learning techniques, while the other teams used classic watermarking approaches. This result leads to the conclusion that an increasing number of people have considered applying deep learning for invisible watermarking.

5.1 Winning Method

The first-place team achieved a final score of 0.8941 against a variety of complex attacks, and the impact on video quality upon embedding is minimal with a PSNR of 43.2. With speeds of 56 FPS and 60 FPS, respectively, the embedding and extraction processes are extremely quick.

This team has designed a hierarchical watermark embedding structure, divided into the following hierarchical levels: 1) One-bit-cube: the video is viewed as a cuboid with width, height, and time as three dimensions, and choose a one-bit-cube as the smallest unit to embed one single bit, 0 or 1. 2) Same-bit adjacent block: Several one-bit-cubes need to form a larger cube to repeatedly place the same bit information as a backup. 3) Watermark package: The adjacent block and the positioning block (used to help find the position of the watermark package in the video) form a large block named watermark package. 4) Channel selection: Considering common color space attacks, the YUV color space of the video is chosen to be embedded, but the Y channel embedding cannot resist brightness and other attacks. And since V channel embedding would cause obvious changes that are sensitive to human eyes, the U channel is finally being chosen.

This local repetition and overall circular structure design can resist blurring attacks and various types of geometric deformation attacks. As long as one watermark package can be recovered, all watermark information can be retrieved. So, a variety of temporal alignment convolution kernels are designed to identify various time deformation attacks during extraction. In addition, team CB used HRnet [9] as the backbone network to detect the attacked video location through the convolutional neural network for the picture-in-picture attack.

5.2 Other Deep Learning Plans

The team securing the fourth position drew inspiration from MBRS [15] to develop their algorithm. Their approach involved an encoder and a decoder component. The encoder aimed to minimize the Mean Squared Error (MSE) Loss and adversarial Loss, ensuring that the watermarked image closely resembled the input image. On the other hand, the decoder focused on minimizing another MSE Loss, striving for the extracted watermark to closely match the encoded watermark. To further enhance the image quality of the watermarked images, they incorporated the LPIPS (Learned Perceptual Image Patch Similarity) Loss [14], ensuring that the watermarked and input images were similar in the feature space. To account for different video resolutions, the team divided each frame into ten patches and embedded the watermark repeatedly across these patches. During watermark decoding, the MBRS decoder was employed to decode each patch individually. If there were a total of L frames, this resulted in a total of $10L$ watermarks. Specifically, to

obtain the final result for a specific bit of the watermark at position k , the team averaged the k th bit across all watermarks. Finally, they binarized the averaged result at 0.5 to derive the k th bit of the watermark.

One of the teams is motivated by RivaGAN [13]. This team employed the same convolutional network layer for the encoder and decoder to produce an attention mask, which is used to identify which pixel region requires attention. The convolutional layer and residual structure were then added for video embedding and watermark extraction. During embedding, for each frame of the video, perform the DWT transformation. Input the resulting low-frequency subband into a convolutional neural network for watermark embedding. Then, perform an inverse DWT transformation on the embedded watermark in the low-frequency subband and combine it with the other mid- to high-frequency subbands of the original frame to generate the watermarked frame. For watermark extraction, perform a DWT transformation on each frame and input the resulting low-frequency subband into a convolutional neural network to extract the watermark information.

There is another team proposed a robust video watermarking algorithm combining Symmetry watermark and Deep Learning (SDL). The SDL architecture includes four modules: 1) Synchronization message embedding module S_{em} , which embeds synchronization message based on symmetry watermark. 2) Watermark embedding module W_{em} , which fuses the cover video frame sequence and the watermark feature to produce the watermarked video frame through the encoder. 3) Synchronization module S_{re} , receiving the attacked watermarked video frame as input. S_{re} extracts the symmetrical peak feature of the video frame and recovers the unknown geometric distortion. 4) Watermark extracting module W_{re} , which receives the synchronized video frames and extracts the watermark message through the decoder.

6 CONCLUSION

In the era of the Internet, video has emerged as the primary medium for information dissemination, posing a significant challenge in terms of copyright protection. Invisible video watermarking has proven to be an effective solution, yet there remain unresolved issues regarding efficient and accurate embedding and extraction of these watermarks. Ensuring minimal impact on video quality during the embedding process is also a crucial consideration for watermarking methods. While traditional techniques continue to dominate the field, the increasing prominence of deep learning in video processing opens new avenues for more reliable watermarking technology that can withstand a broader range of attacks. Consequently, exploring the integration of deep learning into watermarking methods becomes vital. Our invisible video watermark challenge holds immense significance as it not only provides a platform for discussion and comparison of state-of-the-art watermarking algorithms but also draws attention to the critical aspect of multimedia copyright protection.

REFERENCES

- [1] Shehzeen Hussain, Nojan Sheybani, Paarth Neekhara, Xinqiao Zhang¹, and Javier Duarte. 2022. FastStamp: Accelerating Neural Steganography and Digital Watermarking of Images on FPGAs. *arXiv preprint arXiv:2209.12391* (2022).
- [2] Saurav Joshi, Jeetendra Pande, and B. K. Singh. 2018. Watermarking of audio signals using iris data for protecting intellectual property rights of multiple

- owners. *International Journal of Information Technology* 10 (2018), 559–566.
- [3] J. Lee, H. Kim, Y. Yoon, and H. Kim. 2011. DCT-based invisible watermarking for image authentication. *IEEE Transactions on Consumer Electronics* 57, 2 (2011), 539–545.
- [4] Jae-Eun Lee, Young-Ho Seo, and Dong-Wook Kim. 2020. *Convolutional neural network-based digital image watermarking adaptive to the resolution of image and watermark*.
- [5] Y. Liu, S. Tang, R. Liu, L. Zhang, and Z. Ma. 2018. *Secure and robust digital image watermarking scheme using logistic and RSA encryption*. 95–105.
- [6] Rachna Patel, Kalpesh Lad, Mukesh Patel, and Madhavi Desai. 2021. An efficient dct-sbpm based video steganography in compressed domain. *International Journal of Information Technology* 13 (2021), 1073–1078.
- [7] Priya Porwal, Ratnesh N. Chaturvedi, Tanvi Ghag, Nikita Poddar, and Ankita Tawde. 2014. Digital Video Watermarking Using Least Significant Bit (LSB) Technique. *International Journal of Engineering Research & Technology (IJERT)* 3, 1 (January 2014), 2491–2496.
- [8] Sanjana Sinha, Prajnat Bardhan, Swarnali Pramanick, Ankul Jagatramka, Dipak K. Kole, and Aruna Chakraborty. 2011. Digital Video Watermarking using Discrete Wavelet Transform and Principal Component Analysis. *International Journal of Wisdom Based Computing* 1, 2 (August 2011), 7–12.
- [9] Ke Sun, Bin Xiao, Dong Liu, and Jingdong Wang. 2019. Deep High-Resolution Representation Learning for Human Pose Estimation. In *CVPR*.
- [10] Alireza Tavakoli, Zahra Honjani, and Hedieh Sajedi. 2022. Convolutional Neural Network-Based Image Watermarking using Discrete Wavelet Transform. *arXiv preprint arXiv:2210.06179* (2022).
- [11] Yuexin Xiang, Tiantian Li, Wei Ren, and Tianqing Zhu. 2022. Generating Image Adversarial Examples by Embedding Digital Watermarks. *arXiv preprint arXiv:2009.05107* (2022).
- [12] Jin Xun and Jong-Weon Kim. 2018. Robust Digital Watermarking for High-definition Video using Steerable Pyramid Transform, Two Dimensional Fast Fourier Transform and Ensemble Position-based Error Correcting. *KSII Transactions on Internet and Information Systems* 12, 7 (July 2018), 3438–3454. <https://doi.org/10.3837/tiis.2018.07.024>
- [13] Kevin Alex Zhang, Lei Xu, Alfredo Cuesta-Infante, and Kalyan Veeramachaneni. 2019. Robust Invisible Video Watermarking with Attention. (2019). [arXiv:1909.01285](https://arxiv.org/abs/1909.01285) [cs.MM]
- [14] Richard Zhang, Phillip Isola, Alexei A Efros, Eli Shechtman, and Oliver Wang. 2018. The Unreasonable Effectiveness of Deep Features as a Perceptual Metric. In *CVPR*.
- [15] Han Fang Zhaoyang Jia and Weiming Zhang. 2021. MBRS: Enhancing Robustness of DNN-based Watermarking by Mini-Batch of Real and Simulated JPEG Compression. In *arXiv:2108.08211*.